

CS3 Rubric – Restaurant Review Sentiment Analysis Case Study

DS 4002 – Instructor: Karsten Siller; TA: Cole Whittington

Due: 5/6/2026

- **Submission format:** GitHub repository link, PDF, Jupyter Notebook, or Google Colab notebook

Group Assignment (submitted by group leader)

General Description: In this case study, students will use natural language processing (NLP) and machine learning techniques to predict restaurant review sentiment from written customer reviews. Students will explore how text data can be transformed into numerical features and used to classify customer experiences as positive, neutral, or negative.

The goal of this assignment is to introduce students to real-world text classification workflows including preprocessing, feature extraction, supervised learning, and evaluation.

Preparatory Assignments:

- The first ten weeks of class to learn our data project framework
- Previous project cycles (always look to improve from what you learned before)

Why am I doing this?

This project explores how machine learning and natural language processing can be used to predict customer sentiment from restaurant review text. Automated sentiment analysis helps businesses efficiently understand customer experiences and identify patterns in large volumes of online reviews.

Learning Objectives: By completing this case study, students should be able to:

- Clean and preprocess text data
- Convert categorical ratings into sentiment labels
- Apply vectorization methods to text
- Train and evaluate supervised classification models
- Interpret model performance and limitations

- Discuss bias and uncertainty in NLP systems

What am I going to do? Students will build a sentiment classification pipeline using restaurant review text.

Students must:

1. Load and explore the restaurant review dataset
2. Clean and preprocess review text
3. Convert star ratings into sentiment labels
4. Split the dataset into training and testing sets
5. Vectorize review text using TF-IDF or Count Vectorization
6. Train at least two classification models
7. Evaluate model performance using classification metrics
8. Discuss challenges, bias, and uncertainty
9. Present conclusions regarding sentiment prediction performance

How will I know I have succeeded? You will meet expectations on CS3 Restaurant Review Sentiment Analysis Case Study when you follow the criteria in the rubric below.

Spec Category	Spec Details
Formatting	• One PDF report • One notebook file (.ipynb or Colab) • All code used for analysis • Visualizations and evaluation metrics • References in IEEE format ◦ Suggested report length: 3–5 pages
Goal Statement	• One sentence • <u>Goal</u>: Describe the objective of predicting restaurant review sentiment from text • Clearly state how success will be measured using model evaluation metrics such as accuracy or F1-score • Supported by justification paragraphs
Research Question	• One Sentence • <u>Goal</u>: Explain the motivation behind sentiment classification • Craft a question understandable to a general audience

	• Example: Can restaurant review text reliably predict customer sentiment?
Narrative Paragraphs	• Two paragraphs • <u>Goal</u>: Explain why automated sentiment analysis is valuable for businesses and consumers • Discuss the importance of online reviews and NLP applications • Target: “unfamiliar audience” • Include references supporting sentiment analysis and text classification concepts ○ Explain current challenges such as class imbalance and ambiguous language ○ Include a few key details ○ Leave most details to references and guide reader to them
Modeling Approach	• One paragraph • <u>Goal</u>: Describe the NLP and machine learning pipeline used for classification • Include preprocessing, TF-IDF vectorization, train/test split, and classification models • Describe why Logistic Regression and Naive Bayes were selected • Mention evaluation using confusion matrices and weighted metrics
Data Requirements	• Use the Kaggle Restaurant Reviews dataset • Dataset must contain written review text and numerical star ratings • Convert ratings into positive, neutral, and negative sentiment labels
Deliverables	• Cleaned dataset • Jupyter Notebook or Colab notebook • Final written report ○ Visualizations including class distribution and confusion matrices ○ Model comparison and discussion of results
References	• All references should be listed at the end of the document • Use IEEE Documentation style (link)

Acknowledgements: Special thanks to Jess Taggart from UVA CTE for coaching on making this rubric.
This structure is pulled from [Streifer & Palmer \(2020\)](#).